SCALAR FIELDS: EXPANDED ELECTRODYNAMICS

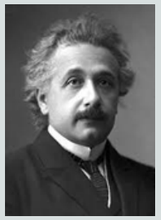
The study of Unidentified Anomalous Phenomena (UAP) presents significant challenges to our current understanding of physics, particularly through the lens of Maxwell's Classical Electrodynamics (CED). Observations of UAP, such as orbs and rods, demonstrate behaviour that defies classical interpretations, including the ability to defy inertia, alter shape at will, and exhibit non-local motion. The absence of the scalar field in Heaviside's revision of Maxwell's equations is a key factor in our inability to fully understand these phenomena. By revisiting Maxwell's original equations and reintroducing the scalar field, we can better explain UAP behaviours and develop technologies that exploit scalar waves for field repulsion, energy manipulation, and trans-medium craft design. A paradigm shift in physics is suggested, integrating a non-local, non-linear approach that harmonizes with Einstein's general relativity, allowing us to explore new realms of interstellar travel.

CLASSICAL ELECTRODYNAMICS

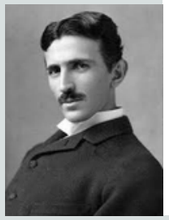
- Maxwells equations refer to rates of change of electromagnetic field quantities.
- Two wave equations are derived from these quantities, one for an electric field vector, and the other for a magnetic field vector.
- **All longitudinal waves parallel to the axis of wave propagation direction are forbidden.**

James Clerk Maxwell: 1831-1879

Oliver Heaviside: 1850-1925



Albert Einstein: 1879-1955



Nikola Tesla: 1856-1943

RELATED LITERATURE

- Morgan, A.D. 2025. Unifying Scalar Fields and Field Repulsion: A New Approach to Understanding Unidentified Anomalous Phenomena and Trans-Medium Travel. February 2025. DOI: 10.13140/RG.2.2.13596.81283
- Morgan, A.D. 2024. Unidentified Anomalous Phenomena (UAP): Orb and Rod Object Orientation and Energy Coupling. November 2024. DOI: 10.13140/RG.2.2.13181.58081
- Maxwell, J. C. 1865. A Treatise on Electricity and Magnetism.
- Heaviside, O. 1889. Electromagnetic Theory.
- Einstein, A. 1915. General Theory of Relativity.
- Tesla, N. 1891. Experiments with Alternate Currents of High Potential and High Frequency.
- Puthoff, H. E. 2004. Gravity Control and Field Propulsion.
- Lück, H. A., & Lück, M. 2008. The Role of Scalar Waves in Electromagnetic Fields and Technology.

AUTHORS

Dr Andrew D. Morgan, March 2025

AFFILIATIONS

Department of Education, Government of Western Australia

1. 3D VECTORS

- A 3D vector can be separated into a circulating component and a back and fourth component.
- Maxwells correct form included the **differential** of the vector potential, which is the magnetic field.
- It also included the **differential** of the vector potential and the scalar potential function, which is the electric field.
- **However, for the electric and magnetic fields, the back and forwards movement along the axis of propagation was not resolved.**

2. HEAVISIDE'S ERROR

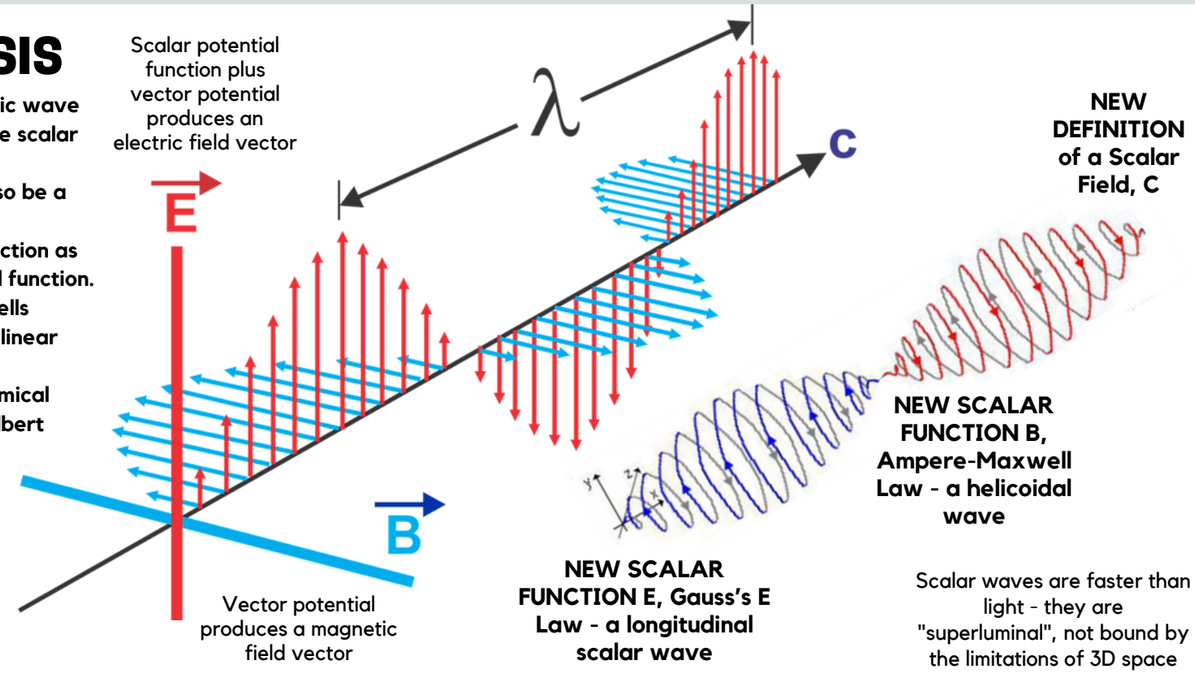
- Heaviside's error in rewriting Maxwell's equations using **vector calculus** was to **not account for the back and forwards movement along the axis of propagation.**
- As a result, until now we have not been able to separate out the longitudinal waves parallel to electromagnetic wave propagation.
- An arbitrary value was assigned for the scalar potential function and set to zero.
- It became the solution to Gauss's law, an unsolvable loop with infinite possibilities.

3. GAUGE THEORY

- This led to gauge theory, and an arbitrary scalar function, which was also a solution in itself to Heaviside alteration.
- Heaviside had inadvertently gauged away the vector potential, along with the electric field and current density.
- The scalar field was set to zero as the Lorenz gauge, but it is actually a non-zero dynamical field.
- All independent works since then have also done this leaving the back and forwards movement unresolved.

KEY TO ANALYSIS

- **Propagation of an electromagnetic wave must contain a 4-gradient of some scalar electrodynamic field.**
- **This electrodynamic field must also be a pressure field.**
- **Any equation must retain this function as a property of the scalar potential function.**
- **It makes the expansion of Maxwells electrodynamics a non-local, non linear theory.**
- **Maxwells expanded electrodynamical theory is then compatible with Albert Einstein's GENERAL RELATIVITY.**



4. THE MATH AND MAXWELLS WORK

- A known electric charge density produces a scalar potential function.
- A known current density produces a vector potential.
- Together, these expression make up what is known; Faradays Law of Induction, Gauss's B Law, and definitions of the vector potential and the scalar potential.
- The electric charge density and scalar potential function, along with the vector. potential produce an **electric field vector.**
- The current density along with the vector potential produce a **magnetic field vector**.

5. THE MATH AND VECTOR POTENTIALS

- The vector potential function is key in determining both the electric and magnetic field vectors.
- Mathematically the vector potential is derived from the current density.
- Gauging it away leaves no way to resolve the longitudinal gradient of scalar wave propagation along the axis of electromagnetic wave propagation.
- Maxwell did not resolve this correctly in his differential equations, and Heaviside then gauged it away (**longitudinal waves were forbidden**).

6. NEW SCALAR QUANTITIES

- In expanded electrodynamics traditional Hertzian transverse EM waves remain unchanged.
- However, an expanded electrodynamical theory is **inclusive of a scalar field**, which resolves smoothly to Maxwell's original classical electrodynamics in limit as C approaches zero.
- Three new scalar waves (**including TESLA WAVES**) are allowed in the expansion of Maxwells electrodynamical equations, these being:
  - A pure scalar wave - the C field,
  - A scalar longitudinal wave - the C field and a longitudinal electric field wave, and
  - A Helicoidal wave - which has the C field, longitudinal electric field wave and the magnetic field wave (a Tesla wave).